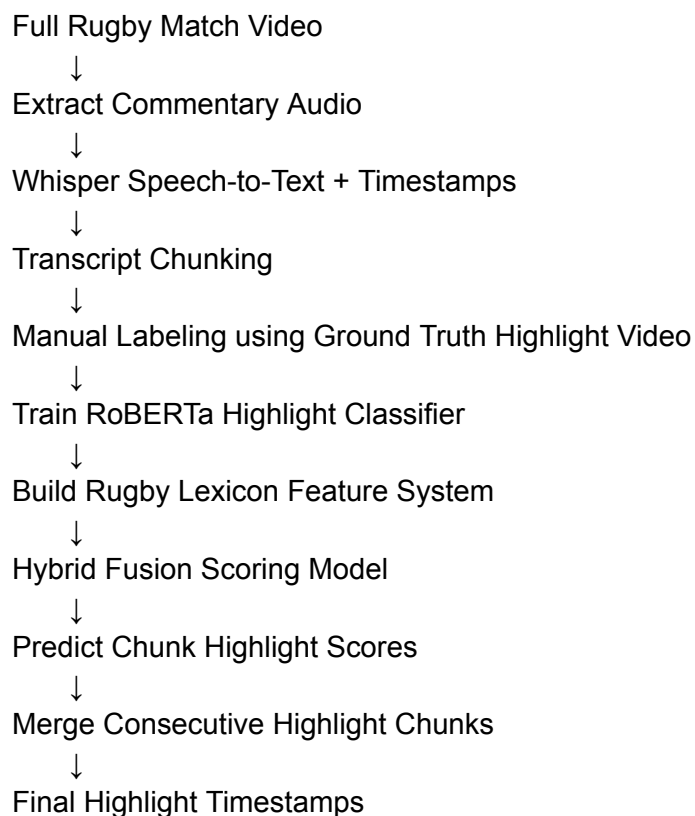


OVERALL SYSTEM FLOW



PHASE 1 — DATA COLLECTION

Step 1: Collect Full Match Videos

What you need:

- Full rugby match videos with commentary audio
- Existing official highlight videos for the same matches

Example:

match_01_full.mp4

match_01_highlights.mp4

match_02_full.mp4
match_02_highlights.mp4

Why highlight videos needed?

They are your **ground truth labels**.

They tell you which moments are real highlights.

PHASE 2 — AUDIO PROCESSING

Step 2: Extract Audio from Full Match Video

Tool: FFmpeg

Command:

```
ffmpeg -i match_01_full.mp4 -vn -acodec pcm_s16le -ar 16000 -ac 1 match_01.wav
```

Output:

match_01.wav

This contains commentary audio.

Step 3: Convert Speech to Text

Tool: Whisper

Command:

```
whisper match_01.wav --model base
```

Output:

[

```
{ "start": 10.2, "end": 13.5, "text": "advantage being played" },  
{ "start": 13.5, "end": 16.9, "text": "he goes wide" },  
{ "start": 16.9, "end": 19.1, "text": "he's over in the corner" }  
]
```

Now you know:

- what was spoken
- exact times

PHASE 3 — CHUNKING

Step 4: Convert Transcript into Chunks

Instead of tiny Whisper segments, make larger windows.

Recommended:

5-second windows with overlap

Example:

0–5 sec
3–8 sec
6–11 sec
9–14 sec

Why overlap?

Because highlights may happen between windows.

Each Chunk Example:

```
{  
  "start": 15,  
  "end": 20,  
  "text": "advantage being played he goes wide"  
}
```

}

PHASE 4 — MANUAL LABELING (VERY IMPORTANT)

Step 5: Create Ground Truth Labels

Use official/manual highlight video.

Example Highlight Video Contains:

16–25 sec try
44–52 sec penalty kick
70–78 sec line break and score

Label Each Chunk

Rule:

If chunk overlaps highlight clip enough → label = 1

Else → label = 0

Example:

Chunk	Time	Label
Chunk A	10–15	0
Chunk B	15–20	1
Chunk C	18–23	1
Chunk D	25–30	0

Final Dataset CSV

start,end,text,label

10,15,"advantage being played",0

15,20,"he goes wide",1

18,23,"he's over in corner",1

25,30,"scrum reset",0

Why this is strong research practice

You are using:

- ✓ Human verified labels
 - ✓ Real highlight ground truth
 - ✓ Supervised learning dataset
-

PHASE 5 — TRAIN RoBERTa MODEL

Step 6: Prepare Text Inputs

Each row:

Text chunk + Label

Example:

"he's over in the corner" → 1

"scrum reset again" → 0

Step 7: Fine-Tune RoBERTa

Tool:

Python + HuggingFace Transformers

Input:

Chunk text

Output:

Highlight probability

Example:

Input:

he breaks through and scores

Output:

0.94 highlight

Result:

RoBERTa learns commentary meaning.

It can detect:

- excitement
 - scoring context
 - attacking pressure
 - semantic patterns
-

 PHASE 6 — BUILD RUGBY LEXICON

Step 8: Create Rugby Dictionary

Category 1: Scoring Terms

try
scores
over
dives
conversion
drop goal

Category 2: Referee Terms

advantage
penalty
offside
forward pass
knock on

Category 3: Set Piece Terms

scrum
lineout
maul
ruck

Category 4: Excitement Terms

brilliant
what a play
huge moment
sensational

PHASE 7 — CONVERT LEXICON TO FEATURES

Step 9: For Each Chunk Count Rugby Signals

Example chunk:

advantage being played he's over

Feature output:

Feature	Value
scoring words	1
referee words	1
set-piece words	0
excitement words	0

Vector:

[1,1,0,0]

PHASE 8 — HYBRID MODEL (MAIN CONTRIBUTION)

Step 10: Combine RoBERTa + Lexicon

For each chunk:

RoBERTa gives:

0.82 probability

Lexicon gives:

0.65 score

Final Hybrid Score

Recommended Formula:

Final Score =
 $0.7 \times \text{RoBERTa Score}$
+
 $0.3 \times \text{Lexicon Score}$

These weights (0.7, 0.3) will be manually adjust to get the most accurate final result. Initial weights are them.

Why 70/30?

RoBERTa understands context better.

Lexicon supports domain clues.

Example:

$$0.7(0.82) + 0.3(0.65) \\ = 0.769$$

Final highlight confidence:

76.9%

PHASE 9 — PREDICT ALL CHUNKS

Step 11: Run Whole Match

Example:

Chunk	Time	Score
A	10–15	0.22
B	15–20	0.77
C	18–23	0.91
D	25–30	0.11

Apply Threshold

Example:

Score > 0.65 = Highlight

Result:

Chunks B and C = highlight

PHASE 10 — MERGE CHUNKS

Step 12: Merge Adjacent Positive Chunks

Example:

15–20 highlight

18–23 highlight

Merge into:

15–23 highlight clip

Optional Padding

Add:

2 sec before

3 sec after

Final:

13–26 clip

Looks better in final video.

PHASE 11 — FINAL OUTPUT

Step 13: Highlight Timestamps

Highlight 1: 13–26 sec

Highlight 2: 44–55 sec

Highlight 3: 70–81 sec

Optional Next Step

Use FFmpeg to auto-cut clips.

PHASE 12 — MODEL EVALUATION (IMPORTANT)

Compare 3 Methods

Method	Precision	Recall	F1
Lexicon Only	0.61	0.52	0.56
RoBERTa Only	0.78	0.74	0.76
Hybrid Model	0.84	0.81	0.82

If Hybrid Best:

You prove research contribution.

FINAL NOVELTY CLAIM

You can state:

A hybrid commentary-based highlight detection model is proposed, combining transformer semantic understanding with rugby-specific lexical scoring using manually labeled highlight data derived from official highlight videos.

BEST IMPLEMENTATION ORDER

Week 1

Collect videos + highlights

Week 2

Whisper transcripts + chunking

Week 3

Manual labeling

Week 4

Train baseline RoBERTa

Week 5

Build lexicon

Week 6

Hybrid scoring

Week 7

Evaluation

Week 8

Report writing

Final Supervisor-Level Explanation

Full-match rugby videos are transcribed using Whisper with timestamps. The transcripts are segmented into overlapping chunks and manually labeled using official highlight videos as ground truth. A RoBERTa model is fine-tuned on this labeled dataset to learn contextual commentary semantics. In parallel, a rugby lexicon extracts domain-specific event cues such as scoring terms, referee decisions, and set-piece terminology. Both outputs are fused in a weighted hybrid scoring model to generate highlight confidence scores. Consecutive positive chunks are merged to produce final highlight timestamps.

My Honest Advice

Your approach is strong, practical, and realistic.

It is much better than trying to invent overly complex deep learning systems with no labeled data.